

ALIVECOR & NEUROBIT: DATA-ACQUISITION STRATEGIES FOR AI IN HEALTHCARE

Ahmed Zahlan, Ravi Prakash Ranjan, and David Hayes wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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In November 2020, in the bustling atmosphere of a conference hall in Boston filled with eager entrepreneurs and industry experts, the chief executive officers (CEOs) of two exceptional healthcare start-ups—AliveCor Inc. (AliveCor) and Neurobit Inc. (Neurobit)—came together for a candid and intriguing discussion. These visionary leaders, who shared a common drive to revolutionize healthcare by integrating artificial intelligence (AI), recounted their unique journeys, challenges, and triumphs in leveraging AI for medical innovation. The two had emerged as innovative healthcare players, commercializing AI-based medical tools. AliveCor, founded in 2011 by Dr. David Albert, Bruce Satchwell, and Kim Barnett, introduced a portable electrocardiogram (ECG) and smartphone app for personal heart monitoring and arrhythmia detection. Neurobit, founded in 2021 by Amiya Patanaik and Kishan, specialized in predictive health analytics and aimed to revolutionize clinical decisions with AI-driven insights based on proprietary data gathered from clinical trials. The start-ups sought to gain an early market-leader advantage by leveraging hard-to-acquire data that would enable the development of robust AI models with high accuracy and set them apart from competitors. Such strategic data ownership enhanced the companies' prospects for securing funding and facilitating early market entry.

However, AliveCor and Neurobit's primary challenge in the early stages was acquiring data—a crucial element for developing competitive and accurate AI models in healthcare. This challenge was not merely a technical hurdle but a strategic dilemma that could define the companies' futures in the industry. AliveCor and Neurobit needed to navigate the complexities of collecting, managing, and leveraging data that complied with regulatory standards while ensuring that their AI models offered high-accuracy insights and value. While they were navigating the data-acquisition process, AliveCor and Neurobit faced other challenges, including those related to securing funding, establishing healthcare collaborations, navigating regulatory landscapes, and competing with other start-ups intent on capturing dominant positions in the same fields. Standing at this critical point in the infancy of their company lifecycles, AliveCor and Neurobit faced a crucial question: What strategy should they use to overcome the challenges of acquiring and utilizing data in a cost-effective and timely way to drive innovation and gain a competitive edge?

ADVANCEMENTS IN AI HEALTHCARE TECHNOLOGY

Kishan and Albert were proud of the work and progress their companies had made in developing AI technologies. AliveCor had been innovating heart-health monitoring systems, utilizing artificial intelligence

for early detection of arrhythmias, and improving cardiac care, reducing costs and hospital visits. Similarly, Neurobit had made significant progress in sleep data analysis and predictive medicine.

The two founders discussed how their technologies joined those of other AI start-up efforts with transformative potential in the healthcare industry. Kishan and Albert emphasized that integrating AI in healthcare boosted the accuracy and efficiency of services and played a crucial role in improving patient outcomes. For example, in cardiology, AI algorithms looked at ECG data to better predict heart diseases, improving early diagnosis. In mental health, AI-powered chatbots administered cognitive behavioural therapy, supporting people with depression or anxiety. Radiology used AI for better image analysis, which helped spot tumours or other issues more accurately. In genomics, AI scanned genetic information to find mutations that caused diseases, opening doors to treatments that were tailored to each patient.

The two founders were confident in the role of AI in shaping the future of healthcare and highlighted its capacity to augment human expertise and provide superior care.

DATA AS THE KEY

While Kishan and Albert agreed on the transformative power of AI in healthcare, they understood that data was crucial to achieving success with their projects and in the marketplace. In healthcare, the success of any AI project depended on having access to accurate and precise data. To complicate matters, the two founders knew that healthcare data came from different sources and formats, and this represented a significant challenge to AI start-ups. To build an effective AI model required considerable expertise and talent to process and make sense of this diverse data. Kishan and Albert emphasized that dealing with data from a variety of sources and being able to analyze it properly were enormous challenges that companies like AliveCor and Neurobit would need to overcome in order to fully tap into AI's power.

SOURCES OF HEALTHCARE DATA

Kishan and Albert reflected on the sources of the data they needed to power their technology. Healthcare data was any information related to health, medical care, or public health, and it came from many different sources:

Research papers: Research papers were valuable data sources for machine learning in healthcare. Medical research papers often contained valuable data from studies and trials, which could be used to train AI models. For example, as reported in medical journals, a model might be trained on patient outcomes data coming from clinical trials on a new medication.

Electronic health records: Most healthcare data used for machine-learning cases¹ were extracted from patient electronic health records (EHR). The data covered a patient's medical history, including diagnoses, medications, lab results, and other clinical information. EHR data was collected and stored by healthcare providers and could be used for disease surveillance and prediction. Machine-learning models were trained on EHR data to identify patterns and predict outcomes.²

¹ IBM described machine learning (ML) as "a branch of artificial intelligence (AI) and computer science that focuses on using data and algorithms to enable AI to imitate the way humans learn, gradually improving its accuracy." IBM, "What is Machine Learning (ML)?," IBM, accessed March 30, 2024, <https://www.ibm.com/topics/machine-learning>.

² Tom Seymour, Dean Frantsovog, and Tod Graeber, "Electronic Health Records (EHR)," *American Journal of Health Sciences (AJHS)* 3, no. 3 (2012): 201–10, <https://doi.org/10.19030/ajhs.v3i3.7139>.

Imaging data: Data scientists extracted imaging data from EHR. Imaging data in healthcare encompassed medical images like X-rays, and computed tomography (CT) and magnetic resonance imaging (MRI) scans, which served diagnostic and treatment planning purposes. Machine-learning algorithms were trained on this imaging data to detect abnormalities and classify images.³ Furthermore, pathological data, such as biopsy results, were used for diagnosis and treatment planning.

Wearable devices: Another source of data was collected from wearable devices and hardware; these became increasingly more pervasive in healthcare due to the fast growth and widespread adoption of smart wearable devices that were capable of recording physiological data virtually in an always-on/always-connected approach.⁴ Wearable sensors gathered information about everyday activities. For example, the freely available University of Southern California human activity dataset (USC-HAD), consisting of well-defined low-level daily activities, was created as a benchmark to help compare different algorithms, particularly those relevant to health-related uses.

In summary, data in healthcare was derived from multiple sources and, through the application of machine learning, enhanced the diagnosis, treatment, and prevention of diseases.

DATA-ACQUISITION MECHANISM

Kishan and Albert explained that they took all sorts of data from different places and formats that weren't ready for AI and organized and prepared it for use by AI in healthcare. They walked through the whole journey of the data—starting from when it was first picked up by devices used in AI models, which they called the extract, transform, load (ETL) process (see Exhibit 1). They also learned that the steps of data acquisition were complicated. These steps included the following:

Data collection: The first step was to collect data from various devices or sources such as sensors, databases, application programming interfaces (APIs), logs, or other data-producing systems. In a healthcare context, data was collected from various sources, such as patient monitoring sensors (e.g., heart-rate monitors and blood-pressure cuffs); EHR systems; medical imaging machines such as MRI and CT scanners; wearable fitness devices (e.g., Fitbit); and even patient surveys. For instance, heart-rate monitors continuously collected and logged patients' heart-rate data.

Data extraction: Once the data was collected, the extraction process began. For example, a start-up that was interested in studying patients with a specific condition like hypertension might extract data by querying the EHR system to retrieve blood-pressure readings and related patient information for individuals diagnosed with hypertension.

Data transformation: After extraction, the data often had to be converted to make it usable for the needed process. This process included cleaning the data, performing calculations or aggregations, and converting data types or formats. In the case of medical notes written by doctors, these unstructured text notes needed to be processed and transformed into structured data. For example, key information like patient symptoms, diagnoses, and treatment recommendations from doctor's notes were extracted and converted into structured fields in the dataset.

³ Ahmed Hosny, Chintan Parmar, John Quackenbush, Lawrence H. Schwartz, and Hugo J. W. L. Aerts, "Artificial Intelligence in Radiology," *Nature Reviews Cancer* 18, no. 8 (2018): 500–510, <https://doi.org/10.1038/s41568-018-0016-5>.

⁴ Farida Sabry, Tamer Eltaras, Wadha Labda, Khawla Alzoubi, and Qutaibah Malluhi, "Machine Learning for Healthcare Wearable Devices: The Big Picture," *Journal of Healthcare Engineering* 2022 (2022): 4653923, <https://doi.org/10.1155/2022/4653923>.

Data loading: The transformed data was then loaded into a target repository such as a database or data warehouse structured patient data, including blood-pressure readings and processed doctor's notes, might have been stored in a healthcare data warehouse during this step.

ETL processes: During the ETL phase, additional processes were applied to ensure data quality and usefulness. For instance, data validation involved checking that blood-pressure readings fell within a reasonable range. Data enrichment involved adding demographic information to the dataset, and deduplication involved identifying and removing any duplicate patient records.

Analytics and reporting: When healthcare data was securely stored and prepared, it became a valuable resource for analytics and reporting. Data scientists were able to use this data to build machine-learning models that predicted patient outcomes, such as the likelihood of cardiovascular events for hypertensive patients. Insights derived from these models guided clinical decisions and improved patient care.

Kishan and Albert shared that moving data through their system was challenging. Data poured in from various places, and each source had its own way of structuring the data. This mix made it challenging to work with and integrate the different types of data they encountered. There was often a large amount of data to manage—especially in big-data scenarios—necessitating robust systems for processing and storage. Sometimes, they had to deal with data as soon as it came in; this added another twist to keeping up with the continuous stream and making sure everything was processed quickly, especially if the data was in real-time processing. Making sure the data was good quality was another challenge. Sometimes they found errors, things that didn't match up, or missing pieces—all of which had to be sorted out while they were transforming and loading the data. As the amount of data and the need for processing it grew, their system also had to grow, and this made scaling up their setup and boosting its performance a tricky affair. Then, there was the challenge of getting different systems, tools, and technologies to work together smoothly within their pipeline, ensuring a seamless data flow from one end to the other. Keeping an eye on the pipeline to immediately catch any errors, breakdowns, or bottlenecks was also crucial to keeping their data reliable.

DATA-ACQUISITION CHALLENGES

Start-ups, especially those in AI healthcare, needed data to make their ideas come to life, but they hit a wall when they looked for specific kinds of patient data. The main issue was that they needed tons of data to create AI solutions that actually worked, but the open-access datasets they could find were too small to meet their needs. To get the AI models to work correctly, they needed to feed them with big chunks of data, and this was a tough task for companies like Neurobit.

Kishan, the CEO of Neurobit, explained:

There was some data out there you can use for free. For example, you might find datasets with thousands of nights' worth of recordings. But that's just a tiny piece of what we need. Many companies can't make it work because that's all they have. These little bits of data are usually given out by some big health organization, like the NIH [National Institutes of Health], and they're just not enough for us to build something that works for everyone.

However, acquiring a sufficient amount of data was just one piece of the puzzle. It was also crucial that the data was good quality and came from a variety of different places and people. This was important in areas like genomics, where it was necessary to understand many variations. For Neurobit, gathering a wide range

of data was key to ensuring that its AI tools were accurate and valuable for everyone. Kishan stressed how important this was:

The core of our data model relies on having diverse data from all over the world. This means we need information that reflects different people, ages, races, health conditions, and genders. This helps us make sure our AI doesn't lean too much one way or another and works well everywhere—be it in India, China, or the US.

Kishan's comment emphasized the importance of having data from all over the world that included different people. This would ensure that the AI solutions were well-rounded and effective. In a world where data was key to producing new ideas, Neurobit and many other companies knew they had to use a wide range of data to stay ahead and succeed everywhere. Neurobit made the most of both freely available and private data from brain research labs, clinical research groups, and healthcare companies all around the globe to improve their AI models. The Neurobit CEO pointed out that, to stand out, "We need a much broader variety of data. Nowadays, it's not just about having a better AI model, but also about having a smart plan to bring all these different pieces of data together." The company was dedicated to collecting a wide variety of data, making them stand out from others. Neurobit knew that the real worth of its product was in the AI model and the special data it learned from—and that this approach made its product unique.

Many companies used their own data, collected from their technology or medical devices, to improve their algorithms. But this method wasn't easy; it took a lot of time to gather enough valuable data, and the companies had to ensure that they followed strict privacy laws. AliveCor was a perfect example. It had started its journey in 2011, but it wasn't until 2020 that it received approval from the US Food and Drug Administration (FDA) to add AI to its product. Gathering and analyzing data while ensuring the company followed all the rules had taken a long time.

The CEO of AliveCor explained:

To get the clinical data we need, we have to follow international guidelines, which is a big and costly task. We need to deal with contracts, ensure we're doing things ethically, and all sorts of other requirements just to collect data for our projects. But we've managed to do it. It's taken a lot of money, time, and hard work, but now we have FDA and other international approvals. Getting to this point wasn't quick. We've been working on this for years.

It wasn't until 2020, nine years after AliveCor was funded, that it could finally offer its AI-powered solution to patients (see Exhibit 2). AliveCor's long journey—from when it first began collecting data to when it was able to launch its AI product—showed the challenges companies could encounter when collecting their own data, especially in areas with strict rules like health care. Its story demonstrated how important it was to be patient, plan carefully, and manage data well to overcome these challenges.

DATA-ACQUISITION STRATEGIES

Kishan highlighted the use of strategic alliances as an approach used by many start-ups, including his own. To build an AI model that helped determine the most appropriate intervention for a particular patient, Neurobit required access to approximately 150 university lab collaborations. This allowed for a comprehensive evaluation and decision-making process. He noted that the company believed it had one of the world's largest collections of polysomnography, or sleep study, data, although it had not publicly claimed this anywhere at this point.

Kishan acknowledged that, while partnerships could be beneficial, they often came with challenges, especially when working with larger organizations. Initiating partnerships with academic institutions or larger companies might have been relatively easy, but maintaining them could be difficult due to the slow, bureaucratic nature of these entities. This sluggishness made data sharing and collaboration time-consuming, posing significant challenges for start-ups accustomed to quick turnaround times. One main challenge, the slow pace of decision-making in larger companies, was often caused by frequent leadership changes and bureaucratic red tape. In contrast, start-ups could have turnaround times as short as 24–48 hours. Scheduling was also difficult; meetings were often scheduled a month in advance due to slow corporate structures. Moreover, while many large enterprises desired to innovate, they often had insufficient incentive. For instance, middle managers in larger organizations might be blamed if an innovative idea failed but also received few rewards when such ideas succeeded. This risk-averse mindset created additional challenges for smaller start-ups seeking fruitful partnerships within an industry.

Neurobit opted for mutually beneficial collaborations in its approach to data collection, working with university research labs, neurology labs, and sleep clinics. Although forming these partnerships could be challenging due to the time and effort involved, the company found that it was crucial to do so, and its primary goal was to save time and effort wherever possible. Neurobit offered low-cost data analysis solutions to these institutions (see Exhibit 3)—a strategy that served two purposes: First, it ensured that Neurobit received anonymous data in return, which it could utilize for its AI model. Second, it allowed the company to bypass the lengthy administrative processes typically associated with universities. Kishan said,

So, most of the people start training on that. So, what we did differently was create an early model. And we started giving it out to researchers, like researchers worldwide. So, that had a different purpose. For example, someone wanted to analyze the respiratory signal, and we were very accurate regarding respiration. Someone wanted to record staging—like, your sleep stages. And we were very accurate in that term. So, we started giving out only those sections to different researchers—sometimes for free or for data exchange; we used to say, “Okay, it will be free if you write our name in as the third-party vendor.” And so, we can exchange data. So that is compensation for us. Or we will charge some money to analyze the data. So, we started the collaboration at the end of 2018 . . . the start of 2019. And until now, we have 150 university partners, 150 . . . labs, and we haven’t claimed it anywhere.

However, funding was often limited in academic settings, and this posed challenges. Neurobit adopted a flexible contracting approach, providing customized solutions to each partner based on their needs. Its company maintained a positive attitude even when requests stretched beyond its capabilities. Instead of declining requests, it agreed to take on the relevant tasks. While this might have seemed like a risky business decision, especially for a small company, Neurobit saw it as an opportunity to adapt and grow. Kishan believed that constantly customizing the company’s services was not a hindrance but a chance to evolve and meet diverse needs.

And there’s lots of roadblocks. We were, like, just small sizes that can fit very well; typically, we never asked for more than \$5,000 because we are also from the academic side, so we know that if you cross \$5,000, it will go through a university admin process and a lot of things. If it is lower than \$5,000, the professor can pay you through the research budget easily. And we don’t want to go through that admin hole and admin process.

Generally, a larger effort expended in collecting data and validating software created a higher barrier for others attempting to duplicate it. This principle held in this case, where the elaborate data collection and model training process added a layer of complexity that would be challenging for others to replicate.

From the outset, AliveCor's terms and conditions had provided the team with anonymous access to ECG data from customers who used their product. AliveCor's team had always maintained a strict privacy policy that ensured they didn't know who the data came from but allowed them to utilize it to enhance their AI capabilities. They had amassed over 180 million ECGs, which recorded the electrical signals in the heart, gathering over a million ECGs into their cloud each week. Albert explained that the company's initial terms and conditions specified that AliveCor would have anonymous access to data from those who bought the company's product but that it wouldn't know the identities of the patients being tested.

This data collection process had started with the launch of the company's first product and continued ever since. AliveCor anticipated reaching the milestone of 200 million ECGs in the cloud in the coming months. This extensive data collection had been instrumental in its AI development, as it had allowed the company to invest significant time, effort, and capital into expert labelling of the ECGs. This meticulous labelling was crucial for building deep learning models, which required a large volume of accurately labelled data. The internal database, considerably larger than any other in this field, enabled the company to construct a superior AI model. While the scale of its operation was huge—it sold three million devices and collected over 180 million ECGs—these figures were not directly proportional: many of AliveCor's devices were used on multiple patients, particularly in clinical settings. As Albert explained,

Our data set is vast and diverse, both geographically and regarding human variability. Additionally, we have longitudinal data, with some users having consistently used an AliveCor device for as long as 10 years. This time-series data is invaluable as it allows us to study the evolution and change in an individual's health over time. Consequently, time is not just a variable in the ECG (which measures voltage over time), but also in our overall data set, where we can analyze ECGs as a function of time.

ALTERNATIVE SOLUTIONS

AliveCor and Neurobit, along with other start-ups in the AI healthcare space, might have considered various alternative approaches to acquiring data to enhance their AI solutions. Kishan and Albert suggested that they had considered other sources of data, which should be considered by other new ventures embarking on AI innovation in healthcare (see Exhibit 4).

Academic collaborations could have offered access to cutting-edge research and a wide network of healthcare data sources. However, such partnerships might not always have been easily attainable, and establishing them might have been time-consuming. Another option was to collect data directly from hospitals and patients through clinical trials; this would have provided direct access to real-world, high-quality healthcare data. This approach, however, would have required a significant financial investment and careful navigation of privacy and consent regulations, and it would have been time-consuming.

Utilizing public and open-source datasets was another potentially cost-effective method of obtaining diverse data for initial model training and testing, despite some issues related to data quality and completeness and a potential mismatch with the start-ups' specific focuses. Forming partnerships with healthcare providers might have offered direct access to clinical data and the potential for long-term collaborations, but negotiating these partnerships could have been complex and time-consuming, necessitating data-sharing agreements.

Purchasing data from third-party vendors was another strategy that could have provided quick access to large datasets, possibly accelerating the development process, but this came with high costs, potential concerns about the relevance and quality of the data, and ethical implications. Additionally, collecting data

from devices it had been selling for years could have allowed the companies to leverage product-usage data that was highly relevant to their goals. This approach, which might have required customer consent and would have been limited by the scale of product deployment, illustrated the diverse challenges and opportunities related to data acquisition for developing AI-driven healthcare solutions.

THE DILEMMA

Kishan and Albert shared stories about how AI healthcare start-ups like theirs had navigated the collection and use of data to train their AI models. They looked at past examples to understand what had worked and what hadn't, especially when identifying the right data for deep machine learning in healthcare. Previously, in the last two decades, the main challenges had been getting ahead by detecting diseases early, diagnosing cancers when they were starting, creating vaccines for future diseases, and predicting new health threats.

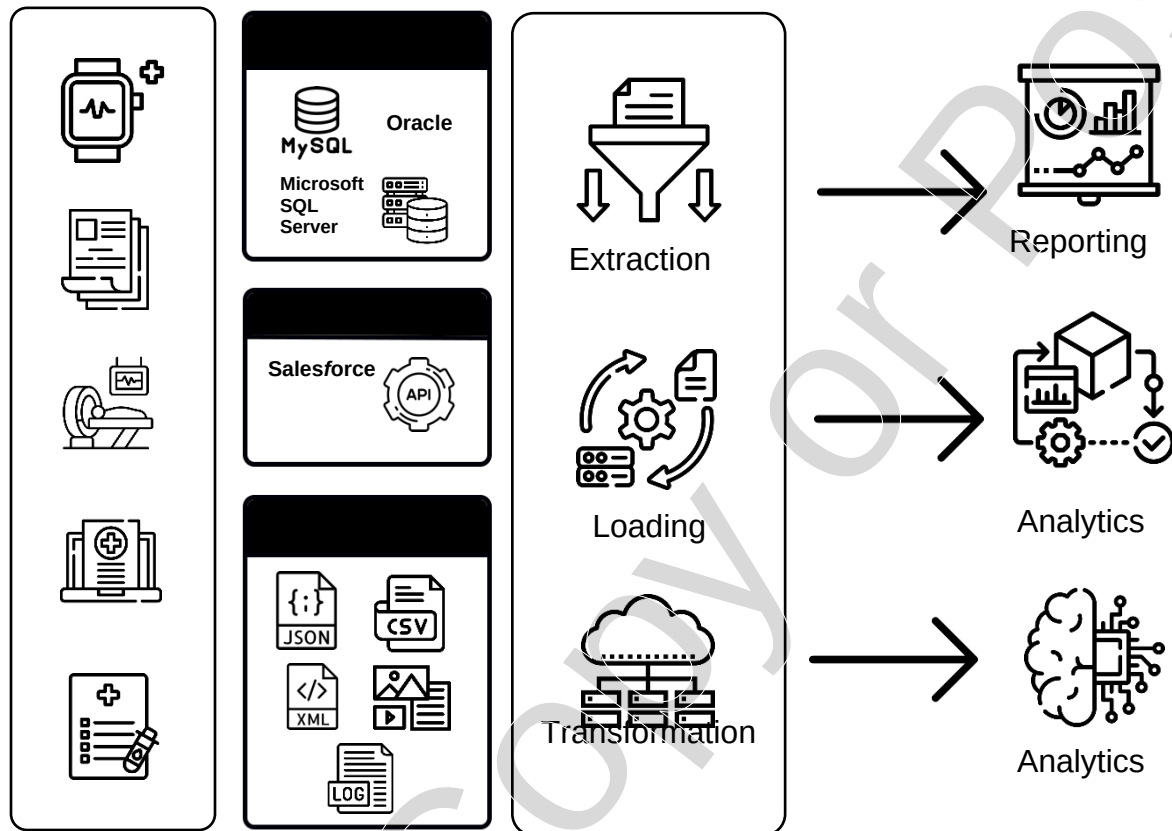
In those days, companies like AliveCor and Neurobit had been lucky. They didn't have much competition early on, and they enjoyed being the first to explore their markets. But AliveCor had spent many years getting to where it was, and this made Albert wonder if new digital healthcare start-ups had had enough time to develop their own AI solutions, as AliveCor and Neurobit had.

Neurobit, especially, depended on working with universities, but as Kishan mentioned, these partnerships weren't easy to initiate and took a lot of time. In the start-up world, time meant money. Meanwhile, other start-ups might have been pouring millions into gathering data directly from hospitals and patients. Yet the founders noticed that venture capitalists hadn't been investing as much in the digital health and AI healthcare sectors lately as they had before—especially in the post-COVID period (see Exhibits 5 and 6).

Data was crucial for innovation in AI healthcare start-ups. Without the correct health data, developing a basic working product was tough. However, to draw in investors and show that their idea could work, the pair needed a proof of concept that used real-world data. This created a Catch-22 scenario: they needed data to move forward, but to get more data, they needed to show some progress.

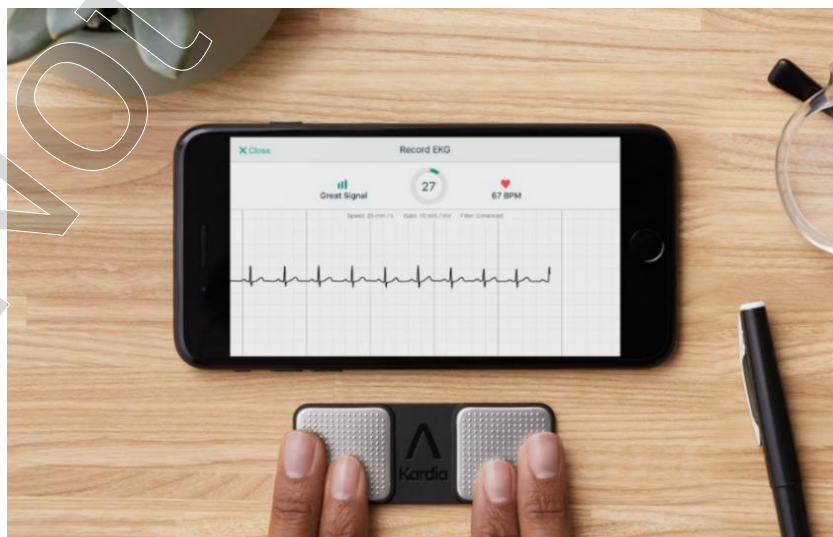
Kishan and Albert pondered this circular problem while coming back to their original dilemma: What strategy should they use to overcome the challenges of acquiring and utilizing data in a cost-effective and timely way to drive innovation and gain a competitive edge?

EXHIBIT 1: DATA PIPELINE TO MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE



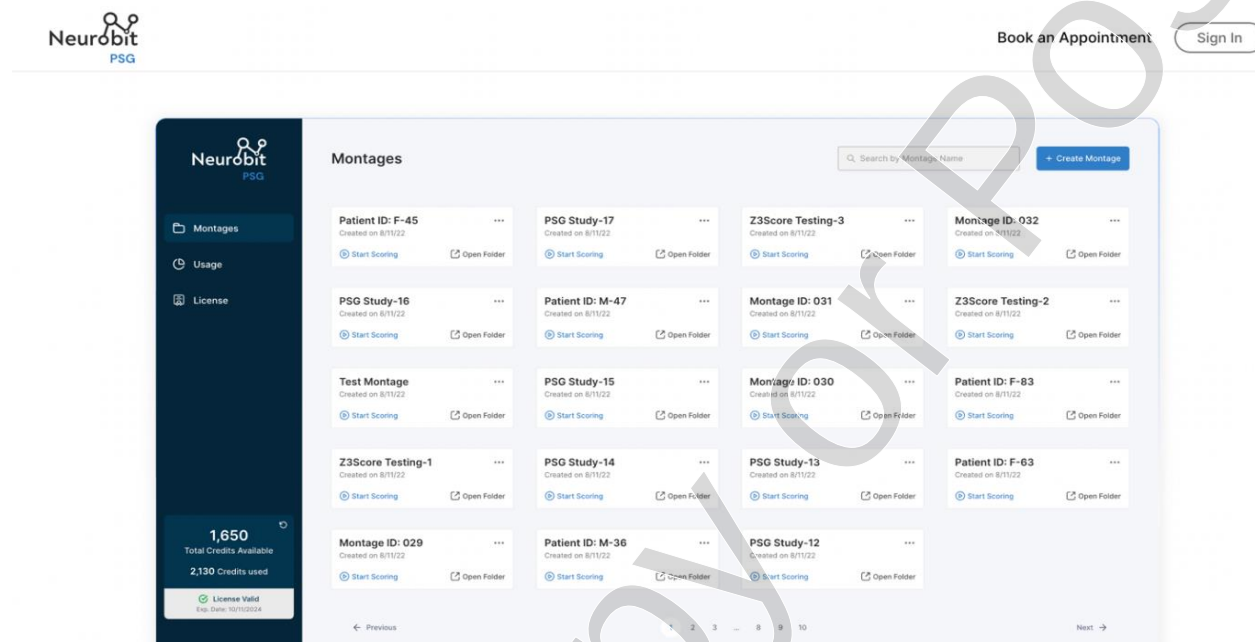
Source: Created by case author.

EXHIBIT 2: ALIVECOR PRODUCT



Source: Company documents.

EXHIBIT 3: NEUROBIT INTERFACE SOLUTION

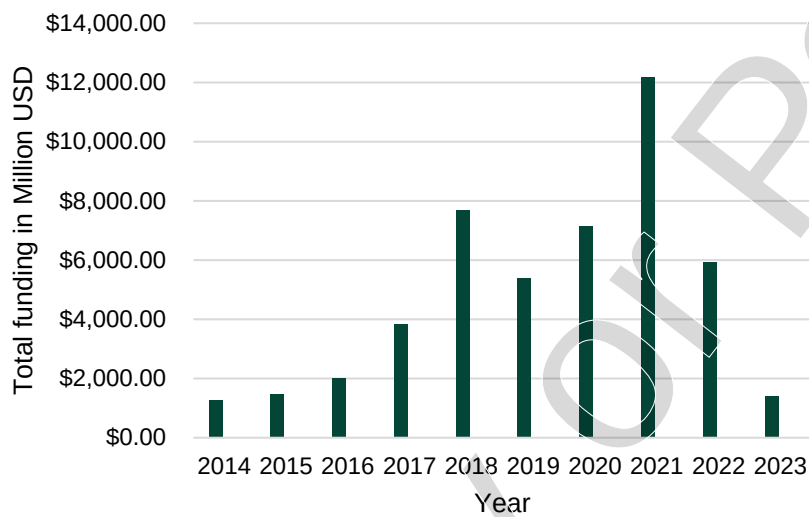


Source: Company documents.

EXHIBIT 4: SUMMARY OF STRATEGIES USED BY AN AI HEALTHCARE START-UP TO ACQUIRE DATA

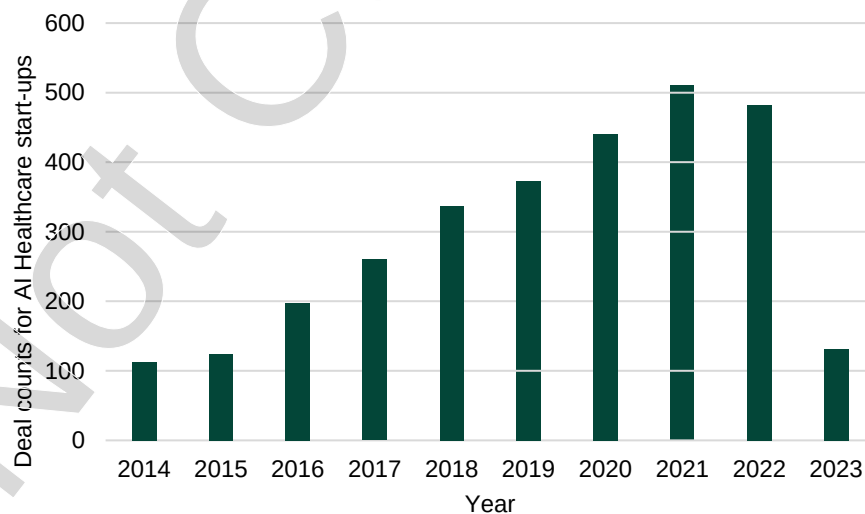
Strategy	Opportunities	Challenges
Academic collaborations	Access to cutting-edge research and a wide network of healthcare data sources	It was time-consuming to establish partnerships; collaborations might not always be easily attainable.
Direct data collection from hospitals and patients (through clinical trials)	Direct access to real-world, high-quality healthcare data	This strategy required significant financial investment; privacy and consent regulations had to be navigated carefully.
Utilization of public and open-source datasets	Cost-effective way to obtain diverse datasets; good for initial model training and testing	Data might not be specific or relevant to the start-up's focus; there might be data quality and completeness issues.
Partnerships with healthcare providers	Direct access to clinical data; potential for long-term collaborations	Negotiating partnerships could be complex and time-consuming; data-sharing agreements were necessary.
Purchasing of data from third-party vendors	Quick access to large datasets; ability to accelerate the development process	This strategy entailed high costs; there were potential issues with data relevance and quality; there were also ethical considerations.
Collection of data from devices sold by the company	Direct leveraging of product-usage data; high relevance and specificity to the company's goals	This strategy might require customer consent and was limited by the scale of product deployment; collecting the data was time-consuming.

Note: AI = artificial intelligence.
Source: Created by case authors

EXHIBIT 5: FUNDING PER YEAR FOR AI HEALTHCARE START-UPS WORLDWIDE

Note: AI = artificial intelligence.

Source: Created by the case author based on C. Sekerak, "The State of Healthcare AI in 5 Charts," CB Insights, August 2, 2023, <https://www.cbinsights.com/research/healthcare-artificial-intelligence-ai-market/>.

EXHIBIT 6: DEALS BETWEEN AI HEALTHCARE START-UPS AND VENTURE CAPITALISTS

Note: AI = artificial intelligence.

Source: Created by the case author based on C. Sekerak, "The State of Healthcare AI in 5 Charts," CB Insights, August 2, 2023, <https://www.cbinsights.com/research/healthcare-artificial-intelligence-ai-market/>.